David Benjamin

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Education

Johns Hopkins University — Baltimore, MD

B.S. in Computer Science | Minor in Entrepreneurship & Management | GPA: 3.94 — *Expected May 2027*

Relevant Coursework: AI-Enabled Software Engineering, Artificial Intelligence, Software System Design, Data Structures, Algorithms, Computer Networks, Computer System Fundamentals, Programming in C/C++ on Linux, Full Stack Development

Experience

SpaceX — September 2026 – December 2026

Software Engineer Intern (Incoming) — *Hawthorne, CA*

Meta — June 2026 – August 2026

Production Engineer Intern — *Burlingame, CA*

- Automated performance tuning for **all production wearables LLMs** by building a Rust CLI that orchestrates safe deployment cloning, controlled load testing, and automated analysis — surfacing configurations that improved throughput by up to **124%**.
- Accelerated serving config discovery from **days to minutes** by designing an adaptive **agent orchestration system** that proposes tuning strategies grounded in live deployment context via read-only tool use, executes experiments, and iterates based on measured results — converging on optimal serving configs from **170+ parameters**.
- Eliminated progress loss across multi-hour tuning experiments and enabled cross-team result sharing by engineering an event-log experiment registry with checkpoint-based resume and queryable trial history.

NASA – National Aeronautics and Space Administration — May 2025 – May 2026

Software Engineer Intern — *Greenbelt, MD*

- Accelerated review cycles for **500+** telescope proposals from **7 days to under 1 hour** by building an AI-powered reviewer assignment system with constraint programming (Google OR-Tools, Django), enforcing 20+ optimization constraints.
- Enabled processing of **2+ hour** optimization workflows by implementing Celery + Redis async architecture with Flower monitoring dashboard for mission-critical telescope proposal reviews.
- Built conflict-detection algorithms (RapidFuzz, Astropy) solving name disambiguation and coordinate matching challenges, ensuring **bias-free reviewer assignments** across **$8 million** in annual telescope funding decisions.
- Sustained uptime for NuSTAR and XRISM scientists during **time-critical review cycles** by orchestrating Kubernetes deployments — authoring manifests (Deployments, Ingress, CronJobs, PVCs) with Kustomize overlays and a GitHub Actions CI/CD pipeline enabling **zero-downtime rolling updates**.

Johns Hopkins Sports Analytics Research Group — May 2024 – May 2025

Lead Software Engineer — *Baltimore, MD*

- Led a **9-person team** to deliver the Atlantic League of Professional Baseball's **first league-wide analytics platform**, providing MLB-caliber data access and tool sharing that modernized team operations.
- Engineered and deployed ETL pipeline (Pandas, AWS Lambda, Docker) and RESTful API (PostgreSQL, API Gateway) democratizing **60K+ Trackman datapoints per game** for **20+ developers** building analytics tools.
- Launched a full-stack web app (Next.js, TypeScript, Express.js, Tailwind CSS) centralizing analytics tools; adopted by **100+ coaches, analysts, and staff** for data-driven decision-making.

Projects & Hackathons

load_in

- Built with a team of 6 an AI developer onboarding engine that transforms any GitHub repo into a personalized curriculum; engineered the external resources agent pipeline — fetching role-tailored YouTube tutorials, articles (Tavily), and reference repos — as a module within a LangGraph **multi-agent system.**
- Grounded AI-generated onboarding quiz answers in actual codebase chunks via **pgvector** retrieval, ensuring source-cited, hallucination-resistant responses.
- Built end-to-end **Slack bot** with an **agentic Q&A handler** — using Claude tool use to query live trainee progress.

Alibaba Global E-Commerce Challenge (2nd / 500+ Global Teams)

- Designed an AR "view in your room" app to reduce purchase uncertainty in Southeast Asian e-commerce — built for Lazada integration using React and Google's AR model-viewer library.
- Placed **2nd** out of **500+ global teams**; pitched live at Alibaba HQ to **70+ engineers and executives**, including C-suite leaders.

Skills & Interests

Languages: Rust, Python, TypeScript, JavaScript, SQL, Bash, C, C++, Java
AI/ML: Vercel AI SDK, pgvector, scikit-learn, Pandas, NumPy, Claude API
Infrastructure: Kubernetes, Docker, AWS, PostgreSQL, Redis, Linux/UNIX, GitHub Actions, Nginx
Activities: HopHacks (Website Team Lead), Pava Accelerator, Climbing, Golf, Skiing, Lacrosse, Phi Gamma Delta